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Inventor(s)	Kojima; Natsuki et al.

Semiconductor light-emitting element

Abstract

A semiconductor light-emitting element includes: an n-type semiconductor layer made of an n-type AlGaN-based semiconductor material; an active layer provided on the n-type semiconductor layer and made of an AlGaN-based semiconductor material; a p-type semiconductor layer provided on the active layer; a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. An Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer.

Inventors: **Kojima; Natsuki (Ishikawa, JP), Inazu; Tetsuhiko (Ishikawa, JP), Niwa; Noritaka (Ishikawa, JP)**

Applicant: **NIKKISO CO., LTD.** (Tokyo, JP)

Family ID: **85038563**

Assignee: **NIKKISO CO., LTD. (Tokyo, JP)**

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Primary Examiner: Manno; Jessica S

Assistant Examiner: Chung; Andrew

Attorney, Agent or Firm: MUNCY, GEISSLER, OLDS & LOWE, P.C.

Background/Summary

RELATED APPLICATION
(1) Priority is claimed to Japanese Patent Application No. 2021-126107, filed on Jul. 30, 2021, the entire content of which is incorporated herein by reference.
BACKGROUND OF THE INVENTION
1. Field of the Invention
(2) The present invention relates to a semiconductor light-emitting element.
2. Description of the Related Art
(3) A semiconductor light-emitting element includes an n-type semiconductor layer, an active layer,

and a p-type semiconductor layer stacked on a substrate. A p-side contact electrode is provided on the p-type semiconductor layer, and a dielectric layer is provided on the p-side contact electrode. In a semiconductor light-emitting element for outputting deep ultraviolet light, rhodium (Rh) having a high reflectivity for the emission wavelength may be used in the p-side contact electrode (see, for example, JP6839320B).
(4) Adhesion of Rh to a dielectric material is poor so that the element reliability may be lowered due to exfoliation of the dielectric layer. Further, when the dielectric layer on the p-side contact electrode is removed to form a connection opening, a damage associated with the removal of the dielectric layer may be caused in the p-side contact electrode exposed in the connection opening with the result that the reflective characteristics of the p-side contact electrode may be lowered.
SUMMARY OF THE INVENTION
(5) The present invention addresses the above-described issues and a purpose thereof is to provide a technology for improving the reliability of a semiconductor light-emitting element.
(6) A semiconductor light-emitting element according to an embodiment of the present invention includes: an n-type semiconductor layer made of an n-type AlGaN-based semiconductor material; an active layer provided on the n-type semiconductor layer and made of an AlGaN-based semiconductor material; a p-type semiconductor layer provided on the active layer; a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. An Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a cross sectional view schematically showing a configuration of a semiconductor light-emitting element according to an embodiment;
(2) FIG. **2** is a cross sectional view schematically showing a configuration of the p-side contact electrode and the p-side current diffusion layer;
(3) FIG. **3** is a cross sectional view schematically showing a configuration of the n-side contact electrode and the n-side current diffusion layer;
(4) FIG. **4** schematically shows a step of manufacturing the semiconductor light-emitting element;
(5) FIG. **5** schematically shows a step of manufacturing the semiconductor light-emitting element;
(6) FIG. **6** schematically shows a step of manufacturing the semiconductor light-emitting element;
(7) FIG. **7** schematically shows a step of manufacturing the semiconductor light-emitting element;
(8) FIG. **8** schematically shows a step of manufacturing the semiconductor light-emitting element; and
(9) FIG. **9** schematically shows a step of manufacturing the semiconductor light-emitting element.
DETAILED DESCRIPTION OF THE INVENTION
(10) The invention will now be described by reference to the preferred embodiments. This does not intend to limit the scope of the present invention, but to exemplify the invention.
(11) A detailed description will be given of an embodiment of the present invention with reference to the drawings. The numerals are used in the description to denote the same elements and a duplicate description is omitted as appropriate. To facilitate the understanding, the relative dimensions of the constituting elements in the drawings do not necessarily mirror the relative dimensions in the actual light-emitting element.
(12) The semiconductor light-emitting element according to the embodiment is configured to emit “deep ultraviolet light” having a central wavelength A approximately equal to or less than 360 nm

and is a so-called deep ultraviolet-light-emitting diode (UV-LED) chip. To output deep ultraviolet light having such a wavelength, an aluminum gallium nitride (AlGaN)-based semiconductor material having a band gap approximately equal to or more than 3.4 eV is used. The embodiment particularly shows a case of emitting deep ultraviolet light having a central wavelength A of about 240 nm-320 nm.

(13) In this specification, the term “AlGaN-based semiconductor material” refers to a semiconductor material containing at least aluminum nitride (AlN) and gallium nitride (GaN) and shall encompass a semiconductor material containing other materials such as indium nitride (InN). Therefore, “AlGaN-based semiconductor materials” as recited in this specification can be represented by a composition In.sub.1-x-yAl.sub.xGa.sub.yN (0<x+y≤1, 0<x<1, 0<y<1). The AlGaN-based semiconductor material shall encompass AlGaN or InAlGaN. The “AlGaN-based semiconductor material” in this specification has a molar fraction of AlN and a molar fraction of GaN equal to or more than 1%, and, preferably, equal to or more than 5%, equal to or more than 10%, or equal to or more than 20%.

(14) Those materials that do not contain AlN may be distinguished by referring to them as “GaN-based semiconductor materials”. “GaN-based semiconductor materials” include GaN or InGaN. Similarly, those materials that do not contain GaN may be distinguished by referring to them as “AlN-based semiconductor materials”. “AlN-based semiconductor materials” include AlN or InAlN.

(15) FIG. **1** is a cross sectional view schematically showing a configuration of a semiconductor light-emitting element **10** according to the embodiment. The semiconductor light-emitting element **10** includes a substrate **20**, a base layer **22**, an n-type semiconductor layer **24**, an active layer **26**, a p-type semiconductor layer **28**, a p-side contact electrode **30**, an n-side contact electrode **32**, a p-side current diffusion layer **34**, an n-side current diffusion layer **36**, a protective layer **38**, a p-side pad electrode **40**, and an n-side pad electrode **42**.

(16) Referring to FIG. **1**, the direction indicated by the arrow A may be referred to as “vertical direction” or “direction of thickness”. Further, the direction away from the substrate **20** may be defined as “upward”, and the direction toward the substrate **20** may be defined as “downward”.

(17) The substrate **20** includes a first principal surface **20**_a_ and a second principal surface **20**_b_ opposite to the first principal surface **20**_a_. The first principal surface **20**_a_ is a crystal growth surface for growing the layers from the base layer **22** to the p-type semiconductor layer **28**. The substrate **20** is made of a material having translucency for the deep ultraviolet light emitted by the semiconductor light-emitting element **10** and is made of, for example, sapphire (Al.sub.2O.sub.3). A fine concave-convex pattern having a submicron (1 μm or less) depth and pitch is formed on the first principal surface **20**_a_. The substrate **20** like this is also called a patterned sapphire substrate (PSS). The second principal surface **20**_b_ is a light extraction substrate for extracting the deep ultraviolet light emitted by the active layer **26** outside. The substrate **20** may be made of AlN or made of AlGaN. The substrate **20** may be an ordinary substrate in which the first principal surface **20**_a_ is comprised of a flat surface that is not patterned.

(18) The base layer **22** is provided on the first principal surface **20**_a_ of the substrate **20**. The base layer **22** is a foundation layer (template layer) to form the n-type semiconductor layer **24**. For example, the base layer **22** is an undoped AlN layer and is, specifically, an AlN (HT-AlN; High Temperature AlN) layer gown at a high temperature. The base layer **22** may further include an undoped AlGaN layer formed on the AlN layer. The base layer **22** may be comprised only of an undoped AlGaN layer when the substrate **20** is an AlN substrate or an AlGaN substrate. In other words, the base layer **22** includes at least one of an undoped AlN layer or an undoped AlGaN layer.

(19) The n-type semiconductor layer **24** is provided on an upper surface **22**_a_ of the base layer **22**. The n-type semiconductor layer **24** is made of an n-type AlGaN-based semiconductor material. For example, the n-type semiconductor layer **24** is doped with Si as an n-type impurity. The composition ratio of the n-type semiconductor layer **24** is selected to transmit the deep ultraviolet

light emitted by the active layer **26**. For example, the n-type semiconductor layer **24** is formed such that the molar fraction of AlN is equal to or more than 25%, and, preferably, equal to or more than 40% or equal to or more than 50%. The n-type semiconductor layer **24** has a band gap larger than the wavelength of the deep ultraviolet light emitted by the active layer **26**. For example, the n-type semiconductor layer **24** is formed to have a band gap equal to or more than 4.3 eV. It is preferable to form the n-type semiconductor layer **24** such that the molar fraction of AlN is equal to or less than 80%, i.e., the band gap is equal to or less than 5.5 eV. It is more desired to form the n-type semiconductor layer **24** such that the molar fraction of AlN is equal to or less than 70% (i.e., the band gap is equal to or less than 5.2 eV). The n-type semiconductor layer **24** has a thickness equal to or more than 1 μm and equal to or less than 3 μm. For example, the n-type semiconductor layer **24** has a thickness of about 2 μm.
(20) The n-type semiconductor layer **24** is formed such that the concentration of Si as the impurity is equal to or more than 1×10.sup.18/cm.sup.3 and equal to or less than 5×10.sup.19/cm.sup.3. It is preferred to form the n-type semiconductor layer **24** such that the Si concentration is equal to or more than 5×10.sup.18/cm.sup.3 and equal to or less than 3×10.sup.19/cm.sup.3 and, more preferably, equal to or more than 7×10.sup.18/cm.sup.3 and equal to or less than 2×10.sup.19/cm.sup.3. In one example, the Si concentration in the n-type semiconductor layer **24** is around 1×10.sup.19/cm.sup.3 and, more specifically, is in a range equal to or more than 8×10.sup.18/cm.sup.3 and equal to or less than 1.5×10.sup.19/cm.sup.3.
(21) The n-type semiconductor layer **24** includes a first upper surface **24***a* and a second upper surface **24***b*. The first upper surface **24***a* is where the active layer **26** is formed, and the second upper surface **24***b* is where the active layer **26** is not formed.
(22) The active layer **26** is provided on the first upper surface **24***a* of the n-type semiconductor layer **24**. The active layer **26** is made of an AlGaN-based semiconductor material and has a double heterojunction structure by being sandwiched between the n-type semiconductor layer **24** and the p-type semiconductor layer **28**. To output deep ultraviolet light having a wavelength equal to or less than 355 nm, the active layer **26** is formed to have a band gap equal to or more than 3.4 eV. For example, the AlN composition ratio of the active layer **26** is selected so as to output deep ultraviolet light having a wavelength equal to or less than 320 nm.
(23) For example, the active layer **26** has a monolayer or multilayer quantum well structure and is comprised of a barrier layer made of an undoped AlGaN-based semiconductor material and a well layer made of an undoped AlGaN-based semiconductor material. The active layer **26** includes, for example, a first barrier layer directly in contact with the n-type semiconductor layer **24** and a first well layer provided on the first barrier layer. One or more pairs of the barrier layer and the well layer may be additionally provided between the first well layer and the p-type semiconductor layer **28**. Each of the barrier layer and the well layer has a thickness equal to or more than 1 nm and equal to or less than 20 nm, and have, for example, a thickness equal to or more than 2 nm and equal to or less than 10 nm.
(24) An electron blocking layer may further be provided between the active layer **26** and the p-type semiconductor layer **28**. The electron blocking layer is made of an undoped AlGaN-based semiconductor material and is formed such that, for example, the molar fraction of AlN is equal to or more than 40%, and, preferably, equal to or more than 50%. The electron blocking layer may be formed such that the molar fraction of AlN is equal to or more than 80% or may be made of an AlN-based semiconductor material that does not contain GaN. The electron blocking layer has a thickness equal to or more than 1 nm and equal to or less than 10 nm. For example, the electron blocking layer has a thickness equal to or more than 2 nm and equal to or less than 5 nm.
(25) The p-type semiconductor layer **28** is formed on the active layer **26**. The p-type semiconductor layer **28** is a p-type AlGaN-based semiconductor material layer or a p-type GaN-based semiconductor material layer. For example, the p-type semiconductor layer **28** is an AlGaN layer or a GaN layer doped with magnesium (Mg) as a p-type impurity. The p-type semiconductor layer **28**

has, for example, a thickness equal to or more than 20 nm and equal to or less than 400 nm.

(26) The p-type semiconductor layer **28** may be comprised of a plurality of layers. The p-type semiconductor layer **28** may include, for example, a p-type clad layer and a p-type contact layer. The p-type clad layer is a p-type AlGaN layer having a relatively high AlN ratio as compared with the p-type contact layer and is provided to be directly in contact with the active layer **26**. The p-type contact layer is a p-type AlGaN layer or a p-type GaN layer having a relatively low AlN ratio as compared with the p-type clad layer. The p-type contact layer is provided on the p-type clad layer and is provided to be directly in contact with the p-side contact electrode **30**. The p-type clad layer may include a p-type first clad layer and a p-type second clad layer.

(27) The composition ratio of the p-type first clad layer is selected to transmit the deep ultraviolet light emitted by the active layer **26**. For example, the p-type first clad layer is formed such that the molar fraction of AlN is equal to or more than 25%, and, preferably, equal to or more than 40% or equal to or more than 50%. The AlN ratio of the p-type first clad layer is, for example, similar to the AlN ratio of the n-type semiconductor layer **24** or larger than the AlN ratio of the n-type semiconductor layer **24**. The AlN ratio of the p-type clad layer may be equal to or more than 70% or equal to or more than 80%. The p-type first clad layer has a thickness equal to or more than 10 nm and equal to or less than 100 nm. For example, the p-type first clad layer has a thickness equal to or more than 15 nm and equal to or less than 70 nm.

(28) The p-type second clad layer is provided on the p-type first clad layer. The p-type second clad layer is a p-type AlGaN layer having a medium AlN ratio and has an AlN ratio lower than that of the p-type first clad layer and higher than that of the p-type contact layer. For example, the p-type second clad layer is formed such that the molar fraction of AlN is equal to or more than 25%, and, preferably, equal to or more than 40% or equal to or more than 50%. The AlN ratio of the p-type second clad layer is configured to be, for example, about ±10% of the AlN ratio of the n-type semiconductor layer **24**. The p-type second clad layer has a thickness equal to or more than 5 nm and equal to or less than 250 nm and has, for example, a thickness equal to or more than 10 nm and equal to or less than 150 nm. The p-type second clad layer may not be provided. The p-type clad layer may be comprised only of the p-type first clad layer.

(29) The p-type contact layer is a p-type AlGaN layer or a p-type GaN layer having a relatively low AlN ratio. The p-type contact layer is formed such that the AlN ratio is equal to or less than 20% in order to obtain proper ohmic contact with the p-side contact electrode **30**. Preferably, the p-type contact layer is formed such that the AlN ratio is equal to or less than 10%, equal to or less than 5%, or 0%. In other words, the p-type contact layer may be made of a p-type GaN-based semiconductor material that does not substantially contain AlN. As a result, the p-type contact layer could absorb the deep ultraviolet light emitted by the active layer **26**. It is preferred to form the p-type contact layer to be thin to reduce the quantity of absorption of the deep ultraviolet light emitted by the active layer **26**. The p-type contact layer has a thickness equal to or more than 5 nm and equal to or less than 30 nm and has, for example, a thickness equal to or more than 10 nm and equal to or less than 20 nm.

(30) The p-side contact electrode **30** is provided on an upper surface **28***a* of the p-type semiconductor layer **28**. The p-side contact electrode **30** can be in ohmic contact with the p-type semiconductor layer **28** (for example, the p-type contact layer) and is made of a material having a high reflectivity for the deep ultraviolet light emitted by the active layer **26**. The p-side contact electrode **30** includes a Rh layer directly in contact with the upper surface **28***a* of the p-type semiconductor layer **28**. The p-side contact electrode **30** may be, for example, comprised only of the Rh layer. The thickness of the Rh layer included in the p-side contact electrode **30** is equal to or more than 50 nm and equal to or less than 200 nm and is, for example, equal to or more than 70 nm and equal to or less than 150 nm.

(31) The film density of the Rh layer included in the p-side contact electrode **30** is equal to or more than 12.0 g/cm.sup.3 and is, for example, equal to or more than 12.2 g/cm.sup.3 and equal to or

less than 12.5 g/cm.sup.3. By configuring the film density of the Rh layer included in the p-side contact electrode **30** to be large, the function of the p-side contact electrode **30** as a reflection electrode can be enhanced. By configuring the film density of the Rh layer to be equal to or more than 12 g/cm.sup.3, the reflectivity equal to or more than 65% for deep ultraviolet light having a wavelength of 280 nm can be obtained. By way of one example, the Rh layer included in the p-side contact electrode **30** has a film density of 12.42 g/cm.sup.3 and has a reflectivity of 66.8% for deep ultraviolet light having a wavelength of 280 nm. Further, the Ar concentration in the Rh layer included in the p-side contact electrode **30** is equal to or more than 1×10.sup.16/cm.sup.3 and less than 1×10.sup.18/cm.sup.3. By configuring the Ar concentration in the Rh layer included in the p-side contact electrode **30** to be low, the film quality of the Rh layer can be improved and the function of the p-side contact electrode **30** as a reflection electrode can be enhanced.
(32) The n-side contact electrode **32** is provided on the second upper surface **24***b* of the n-type semiconductor layer **24**. The n-side contact electrode **32** includes a first Ti layer, an Al layer, a second Ti layer, and a TiN layer stacked successively. The detail of the configuration of the n-side contact electrode **32** will be described separately with reference to FIG. **3**.
(33) The p-side current diffusion layer **34** is provided to be directly in contact with an upper surface **30***a* and a side surface **30***b* of the p-side contact electrode **30** and to cover the entirety of the p-side contact electrode **30**. The p-side current diffusion layer **34** includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. The detail of the configuration of the p-side current diffusion layer **34** will be described separately with reference to FIG. **2**.
(34) The n-side current diffusion layer **36** is provided to cover an upper surface **32***a* and a side surface **32***b* of the n-side contact electrode **32**. The n-side current diffusion layer **36** is configured in a manner similar to the p-side current diffusion layer **34** and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively. The detail of the configuration of the n-side current diffusion layer **36** will be described separately with reference to FIG. **3**.
(35) The protective layer **38** includes a p-side pad opening **38***p* provided on the p-side current diffusion layer **34** and an n-side pad opening **38***n* provided on the n-side current diffusion layer **36**. The protective **38** covers the p-side current diffusion layer **34** in a portion different from the p-side pad opening **38***p* and covers the n-side current diffusion layer **36** in a portion different from the n-side pad opening **38***n*. The protective layer **38** is provided to cover the entirety of the element from above and is provided to cover the upper surface **22***a* of the base layer **22**, the upper surface **24***b* and a side surface **24***c* of the n-type semiconductor layer **24**, a side surface **26***b* of the active layer **26**, the upper surface **28***a* and a side surface **28***b* of the p-type semiconductor layer **28**, the p-side current diffusion layer **34**, and the n-side current diffusion layer **36**.
(36) The protective layer **38** includes a first dielectric layer **44**, a second dielectric layer **46**, and a third dielectric layer **48**. Each of the first dielectric layer **44**, the second dielectric layer **46**, and the third dielectric layer **48** is made of a material that does not substantially absorb the deep ultraviolet light emitted by the active layer **26** and is made of a material having a transmittance equal to or more than 80% for the wavelength of the deep ultraviolet light emitted by the active layer **26**. Such a material is exemplified by an oxide material such as silicon oxide (SiO.sub.2), aluminum oxide (Al.sub.2O.sub.3), and hafnium oxide (HfO.sub.2).
(37) The first dielectric layer **44** is directly in contact with the n-type semiconductor layer **24**, the active layer **26**, the p-type semiconductor layer **28**, the p-side current diffusion layer **34**, and the n-side current diffusion layer **36**. The first dielectric layer **44** is made of a first oxide material and is made of SiO.sub.2, Al.sub.2O.sub.3, or HfO.sub.2. The first dielectric layer **44** is preferably made of SiO.sub.2. The thickness of the first dielectric layer **44** is equal to or more than 300 nm and equal to or less than 1500 nm and is, for example, about equal to more than 600 nm and equal to or less than 1000 nm.
(38) The second dielectric layer **46** is provided on the first dielectric layer **44** and is provided to cover the entirety of the first dielectric layer **44**. The second dielectric layer **46** is made of a second

oxide material different from that of the first dielectric layer **44** and is made of SiO.sub.2, Al.sub.2O.sub.3, or HfO.sub.2. The second dielectric layer **46** is preferably made of Al.sub.2O.sub.3. By configuring the material of the second dielectric layer **46** to be different from the material of the first dielectric layer **44**, pin holes that could be produced in the first dielectric layer **44** can be blocked to increase the quality of sealing. The thickness of the second dielectric layer **46** is equal to or more than 10 nm and equal to or less than 100 nm and is, for example, equal to or more than 20 nm and equal to or less than 50 nm.
(39) The third dielectric layer **48** is provided on the second dielectric layer **46** and is provided to cover the entirety of the second dielectric layer **46**. The third dielectric layer **48** is made of a third oxide material different from the second oxide material forming the second dielectric layer **46** and is preferably made of SiO.sub.2. By configuring the material of the third dielectric layer **48** to be different from the material of the second dielectric layer **46**, pin holes that could be produced in the second dielectric layer **46** can be blocked to increase the sealing performance. The thickness of the third dielectric layer **48** is equal to or more than 10 nm and equal to or less than 100 nm and is, for example, equal to or more than 20 nm and equal to or less than 50 nm.
(40) The p-side pad electrode **40** is provided on the p-side current diffusion layer **34** and connects to the p-side current diffusion layer **34** in the p-side pad opening **38***p*. The p-side pad electrode **40** is provided to block the p-side pad opening **38***p* and overlaps the protective layer **38** outside the p-side pad opening **38***p*. The p-side pad electrode **40** is electrically connected to the p-side contact electrode **30** via the p-side current diffusion layer **34**.
(41) The n-side pad electrode **42** is provided on the n-side current diffusion layer **36** and connects to the n-side current diffusion layer **36** in the n-side pad opening **38***n*. The n-side pad electrode **42** is provided to block the n-side pad opening **38***n* and overlaps the protective layer **38** outside the n-side pad opening **38***n*. The n-side pad electrode **42** is electrically connected to the n-side contact electrode **32** via the n-side current diffusion layer **36**.
(42) The p-side pad electrode **40** and the n-side pad electrode **42** are portions bonded when the semiconductor light-emitting element **10** is mounted on a package substrate or the like. The p-side pad electrode **40** and the n-side pad electrode **42** include, for example, a Ni/Au, Ti/Au, or Ti/Pt/Au stack structure. The thickness of each of the p-side pad electrode **40** and the n-side pad electrode **42** is equal to or more than 100 nm and is, for example, equal to or more than 200 nm and equal to or less than 1000 nm.
(43) FIG. **2** schematically shows a configuration of the p-side contact electrode **30** and the p-side current diffusion layer **34**. The p-side contact electrode **30** is comprised of a Rh layer. The p-side current diffusion layer **34** includes a first TiN layer **50**, a multilayer metal film **52**, and a second TiN layer **54**. The p-side current diffusion layer **34** may further include a Ti layer **56** and an Au layer **58**.
(44) The first TiN layer **50** of the p-side current diffusion layer **34** is directly in contact with the Rh layer of the p-side contact electrode **30**. The multilayer metal film **52** of the p-side current diffusion layer **34** is provided on the first TiN layer **50**. The second TiN layer **54** of the p-side current diffusion layer **34** is provided on the multilayer metal film **52**. The first TiN layer **50** and the second TiN layer **54** are made of conductive TiN. The thickness of each of the first TiN layer **50** and the second TiN layer **54** is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 50 nm and equal to or less than 150 nm.
(45) The multilayer metal film **52** of the p-side current diffusion layer **34** includes a Ti layer **52***a* and a Rh layer **52***b*. The multilayer metal film **52** may include a plurality of Ti layers **52***a* and a plurality of Rh layers **52***b* that are alternately stacked. The thickness of each of the Ti layer **52***a* and the Rh layer **52***b* is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 20 nm and equal to or less than 150 nm. The film density of the Rh layer **52***b* included in the p-side current diffusion layer **34** is equal to or more than 12.0 g/cm.sup.3 and is, for example, equal to or more than 12.2 g/cm.sup.3 and less than 12.5 g/cm.sup.3. The film

density of the Rh layer included in the p-side contact electrode **30** may be larger than the film density of the Rh layer **52***b*. By way of one example, the Rh layer **52***b* included in the p-side current diffusion layer **34** has a film density of 12.37 g/cm.sup.3 and has a reflectivity of 66.0% for ultraviolet light having a wavelength of 280 nm. The Ar concentration in the Rh layer **52***b* included in the p-side current diffusion layer **34** is larger than the Ar concentration in the Rh layer included in the p-side contact electrode **30**. The Ar concentration in the Rh layer **52***b* is, for example, equal to or more than 1×10.sup.18/cm.sup.3 and equal to or less than 5×10.sup.20/cm.sup.3.
(46) The Ti layer **56** of the p-side current diffusion layer **34** is provided on the second TiN layer **54**. The thickness of the Ti layer **56** is equal to or more than 1 nm and equal to or less than 50 nm and is, for example, equal to or more than 5 nm and equal to or less than 25 nm. The Au layer **58** of the p-side current diffusion layer **34** is provided on the Ti layer **56**. The thickness of the Au layer **58** is equal to or more than 100 nm and equal to or less than 500 nm and is, for example, equal to or more than 150 nm and equal to or less than 300 nm.
(47) FIG. **3** schematically shows a configuration of the n-side contact electrode **32** and the n-side current diffusion layer **36**. The n-side contact electrode **32** includes a first Ti layer **60**, an Al layer **62**, a second Ti layer **64**, and a TiN layer **66**. The n-side current diffusion layer **36** includes a first TiN layer **68**, a multilayer metal film **70**, and a second TiN layer **72**. The n-side current diffusion layer **36** may further include a Ti layer **74** and an Au layer **76**.
(48) The first Ti layer **60** of the n-side contact electrode **32** is directly in contact with the second upper surface **24***b* of the n-type semiconductor layer **24**. The thickness of the first Ti layer **60** is equal to or more than 1 nm and equal to or less than 10 nm and is, preferably, equal to or less than 5 nm or equal to or less than 2 nm. The Al layer **62** of the n-side contact electrode **32** is provided on the first Ti layer **60** and is directly in contact with the first Ti layer **60**. The thickness of the Al layer **62** is equal to or more than 200 nm and is, for example, equal to or more than 300 nm and equal to or less than 1000 nm. The second Ti layer **66** of the n-side contact electrode **32** is provided on the Al layer **62** and is directly in contact with the Al layer **62**. The thickness of the second Ti layer **64** is equal to or more than 1 nm and equal to or less than 50 nm and is, for example, equal to or more than 5 nm and equal to or less than 25 nm. The TiN layer **66** of the n-side contact electrode **32** is provided on the second Ti layer **64** and is directly in contact with the second Ti layer **64**. The TiN layer **66** is made of conductive TiN. The thickness of the TiN layer **66** is equal to or more than 5 nm and equal to or less than 100 nm and is, for example, equal to or more than 10 nm and equal to or less than 50 nm.
(49) The first TiN layer **68** of the n-side current diffusion layer **36** is directly in contact with the upper surface **32***a* and the side surface **32***b* of the n-side contact electrode **32**. The multilayer metal film **70** of the n-side current diffusion layer **36** is provided on the first TiN layer **68**. The second TiN layer **72** of the n-side current diffusion layer **36** is provided on the multilayer metal film **70**. The first TiN layer **68** and the second TiN layer **72** are made of conductive TiN. The thickness of each of the first TiN layer **68** and the second TiN layer **72** is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 50 nm and equal to or less than 150 nm.
(50) Like the multilayer metal film **52** of the p-side current diffusion layer **34**, the multilayer metal film **70** of the n-side current diffusion layer **36** is comprised of a Ti layer **70***a* and a Rh layer **70***b*. The multilayer metal film **70** may include a plurality of Ti layers **70***a* and a plurality of Rh layers **70***b* that are alternately stacked. The thickness of each of the Ti layer **70***a* and the Rh layer **70***b* included in the multilayer metal film **70** is equal to or more than 10 nm and equal to or less than 200 nm and is, for example, equal to or more than 20 nm and equal to or less than 150 nm. The film density and the Ar concentration of the Rh layer **70***b* included in the n-side current diffusion layer **36** are similar to the film density and the Ar concentration of the Rh layer **52***b* included in the p-side current diffusion layer **34**. The film density of the Rh layer **70***b* is equal to or more than than 12.0 g/cm.sup.3 and is, for example, equal to or more than 12.2 g/cm.sup.3 and equal to or less

than 12.5 g/cm.sup.3. The Ar concentration in the Rh layer **70***b* is equal to or more than 1×10.sup.18/cm.sup.3 and less than 1×10.sup.21/cm.sup.3.
(51) The Ti layer **74** of the n-side current diffusion layer **36** is provided on the second TiN layer **72**. The thickness of the Ti layer **74** is equal to or more than 1 nm and equal to or less than 50 nm and is, for example, equal to or more than 5 nm and equal to or less than 25 nm. The Au layer **76** of the n-side current diffusion layer **36** is provided on the Ti layer **74**. The thickness of the Au layer **76** is equal to or more than 100 nm and equal to or less than 500 nm and is, for example, equal to or more than 150 nm and equal to or less than 300 nm.
(52) A description will now be given of a method of manufacturing the semiconductor light-emitting element **10**. FIGS. **4**-**9** schematically show steps of manufacturing the semiconductor light-emitting element **10**. First, referring to FIG. **4**, the base layer **22**, the n-type semiconductor layer **24**, the active layer **26**, and the p-type semiconductor layer **28** are formed on the first principal surface **20***a* of the substrate **20** sequentially.
(53) The substrate **20** is, for example, a patterned sapphire substrate. The base layer **22** includes, for example, an HT-AlN layer and an undoped AlGaN layer. The n-type semiconductor layer **24**, the active layer **26**, and the p-type semiconductor layer **28** are semiconductor layers made of an AlGaN-based semiconductor material, an AlN-based semiconductor material, or a GaN-based semiconductor material and can be formed by a well-known epitaxial growth method such as the metal organic vapor phase epitaxy (MOVPE) method and the molecular beam epitaxy (MBE) method.
(54) Subsequently, as shown in FIG. **4**, a mask **80** is formed on the upper surface **28***a* of the p-type semiconductor layer **28** by using, for example, a publicly known lithographic technology. The p-type semiconductor layer **28** and the active layer **26** in a region not overlapping the mask **80** are removed by dry-etching or the like while the mask **80** is being formed, to expose the second upper surface **24***b* of the n-type semiconductor layer **24**. This etching step forms the side surfaces **28***b* of the p-type semiconductor layer **28**, the side surface **26***b* of the active layer **26**, and the second upper surface **24***b* of the n-type semiconductor layer **24**. The mask **80** is then removed.
(55) Then, as shown in FIG. **5**, the p-side contact electrode **30** is formed on the upper surface **28***a* of the p-type semiconductor layer **28** by using, for example, a publicly known lithographic technology. The p-side contact electrode **30** includes a Rh layer directly in contact with the upper surface **28***a* of the p-type semiconductor layer **28**. The Rh layer of the p-side contact electrode **30** is formed by deposition at a temperature equal to or less than 100° C. By forming the Rh layer by deposition, the damage to the upper surface **28***a* of the p-type semiconductor layer **28** can be reduced and the contact resistance of the p-side contact electrode **30** can be improved as compared with the case of using sputtering.
(56) After the p-side contact electrode **30** is formed, the p-side contact electrode **30** is annealed. The p-side contact electrode **30** is annealed by using, for example, the rapid thermal annealing (RTA) method at a temperature equal to or more than 500° C. and equal to or less than 650° C. The annealing process of the p-side contact electrode **30** lowers the contact resistance of the p-side contact electrode **30** and increases the film density of the Rh layer included in the p-side contact electrode **30** to be equal to or more than 12 g/cm.sup.3. The film density of the Rh layer formed by deposition at a temperature equal to or less than 100° C. is less than 12 g/cm.sup.3 and is, for example, equal to or more than 11.6 g/cm.sup.3 and equal to or less than 11.9 g/cm.sup.3. Further, the reflectivity of the Rh layer formed by deposition at a temperature equal to or less than 100° C. for the wavelength of 280 nm is less than 65% and is, for example, about 60%-61%. Meanwhile, the film density of the Rh layer of the p-side contact electrode **30** after the annealing process will be, for example, equal to or more than 12.2 g/cm.sup.3 and equal to or less than 12.5 g/cm.sup.3. The reflectivity of the Rh layer of the p-side contact electrode **30** for the wavelength of 280 nm after the annealing process is equal to or more than 65% and is, for example, 66.8%.
(57) Then, as shown in FIG. **6**, the n-side contact electrode **32** is formed on the second upper

surface **24***b* of the n-type semiconductor layer **24** by using, for example, a publicly known lithographic technology. The n-side contact electrode **32** is in contact with the second upper surface **24***b* of the n-type semiconductor layer **24** and includes the first Ti layer **60**, the Al layer **62**, the second Ti layer **64**, and the TiN layer **66** (see FIG. **3**) stacked successively. The first Ti layer **60**, the Al layer **62**, the second Ti layer **64**, and the TiN layer **66** forming the n-side contact electrode **32** are formed by sputtering.

(58) After the n-side contact electrode **32** is formed, the n-side contact electrode **32** is annealed. The n-side contact electrode **32** is annealed by using, for example, the RTA method at a temperature equal to or more than 500° C. and equal to or less than 650° C. The annealing process of the n-side contact electrode **32** lowers the contact resistance of the n-side contact electrode **32**.

(59) Subsequently, as shown in FIG. **7**, the p-side current diffusion layer **34** is formed to cover the upper surface **30***a* and the side surface **30***b* of the p-side contact electrode **30**, and the n-side current diffusion layer **36** is formed to cover the upper surface **32***a* and the side surface **32***b* of the n-side contact electrode **30**, by using, for example, a publicly known lithographic technology. The p-side current diffusion layer **34** and the n-side current diffusion layer **36** can be formed simultaneously at a temperature equal to or less than 100° C. by sputtering that uses an Ar gas. The p-side current diffusion layer **34** and the n-side current diffusion layer **36** may be formed separately. As shown in FIGS. **2** and **3**, the first TiN layers **50**, **68** are formed first, and then the multilayer metal films **52**, **70** including the Ti layers **52***a*, **70***a* and the Rh layers **52***b*, **70***b* are formed on the first TiN layers **50**, **68**, and the second TiN layers **54**, **72** are formed on the multilayer metal films **52**, **70**. The Ti layers **56**, **74** and the Au layers **58**, **76** may further be formed on the second TiN layers **54**, **72**. The film density of the Rh layer formed by sputtering at a temperature equal to or less than 100° C. is less than 12.0 g/cm.sup.3 and is, for example, equal to or more than 11.6 g/cm.sup.3 and equal to or less than 11.9 g/cm.sup.3. The reflectivity of the Rh layer formed by sputtering at a temperature equal to or less than 100° C. for the wavelength of 280 nm is less than 65% and is, for example, about 60%-61%.

(60) Subsequently, as shown in FIG. **8**, a mask **82** is formed on the n-side semiconductor layer **24**, the active layer **26**, the p-type semiconductor layer **28**, the p-side current diffusion layer **34**, and the n-side current diffusion layer **36** by using, for example, a publicly known lithographic technology. After the mask **82** is formed, the n-type semiconductor layer **24** in a region not overlapping the mask **82** is removed by dry-etching or the like to expose the upper surface **22***a* of the base layer **22**. This etching step forms the side surface **24***c* of the n-type semiconductor layer **24**. The mask **82** is then removed.

(61) Subsequently, as shown in FIG. **9**, the protective layer **38** is formed to cover the entirety of the upper surface of the element structure. First, the first dielectric layer **44** made of the first oxide material is formed. The first dielectric layer **44** can be made of SiO.sub.2 and can be formed by plasma enhanced chemical vapor deposition (PECVD). The first dielectric layer **44** is provided to cover the upper surface **22***b* of base layer **22**, the second upper surface **24***b* and the side surface **24***c* of the n-type semiconductor layer **24**, a side surface **26***c* of the active layer **26**, the upper surface **28***a* and a side surface **28***c* of the p-type semiconductor layer **28**, the p-side current diffusion layer **34**, and the n-side current diffusion layer **36**.

(62) Subsequently, the second dielectric layer **46** made of the second oxide material is formed on the first dielectric layer **44**. The second dielectric layer **46** is formed to cover the entirety of the upper surface of the first dielectric layer **44**. The second dielectric layer **46** can be made of Al.sub.2O.sub.3 and can be formed by atomic layer deposition (ALD). By using the ALD method, a tight dielectric film having a high film density can be formed. The third dielectric layer **48** made of SiO.sub.2 is then formed on the second dielectric layer **46**. The third dielectric layer **48** is formed to cover the entirety of the upper surface of the second dielectric layer **46**. The third dielectric layer **48** can be formed by using the ALD method.

(63) In the step of forming the protective layer **38**, the p-side current diffusion layer **34** and the n-

side current diffusion layer **36** are heated at a temperature equal to or more than 200° C. and equal to or less than 400° C. The film density of the Rh layers **52***b*, **70***b* included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** is increased to be equal to or more than 12.0 g/cm.sup.3 by heating the p-side current diffusion layer **34** and the n-side current diffusion layer **36** at a temperature equal to or more than 200° C. and equal to or less than 400° C. The film density of the Rh layer formed by sputtering at a temperature equal to or less than 100° C. is less than 12.0 g/cm.sup.3 and is, for example, equal to or more than 11.6 g/cm.sup.3 and equal to or less than 11.9 g/cm.sup.3. Further, the reflectivity of the Rh layers **52***b*, **70***b* formed by sputtering at a temperature equal to or less than 100° C. for ultraviolet light having a wavelength of 280 nm is less than 65% and is, for example, about 60%-61%. Meanwhile, the Rh layers **52***b*, **70***b* heated at a temperature equal to or more than 200° C. and equal to or less than 400° C. has, for example, a film density equal to or more than 12.2 g/cm.sup.3 and less than 12.5 g/cm.sup.3 and has a reflectivity equal to or more than 65% (for example, 66%) for ultraviolet light having a wavelength of 280 nm.
(64) When the Rh layer formed by sputtering at a temperature equal to or less than 100° C. is annealed at a temperature equal to or more than 500° C. and equal to or less than 650° C., the film density of the annealed Rh layer will be less than 12.0 g/cm.sup.3. It is therefore preferred to anneal the p-side contact electrode **30** and the n-side contact electrode **32** before the p-side current diffusion layer **34** and the n-side current diffusion layer **36** are formed.
(65) Subsequently, as shown in FIG. **1**, the protective layer **38** is removed in part by dry-etching or the like to form the p-side pad opening **38***p* and the n-side pad opening **38***n*, by using, for example, a publicly known lithographic technology. The p-side pad opening **38***p* and the n-side pad opening **38***n* are formed to extend through the first dielectric layer **44**, the second dielectric layer **46**, and the third dielectric layer **48**. The p-side current diffusion layer **34** is exposed in the p-side pad opening **38***p*, and the n-side current diffusion layer **36** is exposed in the n-side pad opening **38***n*. Subsequently, the p-side pad electrode **40** connected to the p-side current diffusion layer **34** in the p-side pad opening **38***p* is formed to block the p-side pad opening **38***p*, and the n-side pad electrode **42** connected to the n-side current diffusion layer **36** in the n-side pad opening **38***n* is formed to block the n-side pad opening **38***n*. The p-side pad electrode **40** and the n-side pad electrode **42** may be formed simultaneously but may be formed separately.
(66) The semiconductor light-emitting element **10** shown in FIG. **1** is completed through the steps described above.
(67) According to the embodiment, adhesion of the protective layer **38** to the p-side contact electrode **30** can be improved by covering the upper surface **30***a* and the side surface **30***b* of the p-side contact electrode **30**, which includes the Rh layer, and forming the protective layer **38** in contact with the p-side current diffusion layer **34**.
(68) According to the embodiment, the Rh layer included in the p-side current diffusion layer **34** can be used as an etching stop layer when the pad opening **38***p* is formed in the protective layer **38**. This prevents a damage to the Rh layer included in the p-side contact electrode **30** and prevents the reflective characteristics of the p-side contact electrode **30** from being lowered.
(69) According to the embodiment, the film density of the Rh layer included in the p-side contact electrode **30** is equal to or more than than 12.0 g/cm.sup.3 so that the Rh layer included in the p-side contact electrode **30** can be configured to be dense to increase the reflectivity for ultraviolet light. Further, the film density of the Rh layer **52***b* included in the p-side current diffusion layer **34** is also equal to or more than 12.0 g/cm.sup.3 so that the Rh layer included in the p-side currently diffusion layer **34** can be configured to be dense to increase the quality of sealing by the p-side current diffusion layer **34**.
(70) According to the embodiment, the Rh layer included in the p-side contact electrode **30** can be configured to be dense and the reflectivity for ultraviolet light can be increased by configuring the film density of the Rh layer included in the p-side contact electrode **30** to be larger than the film density of the Rh layer **52***b* included in the p-side current diffusion layer **34** and, for example, equal

to or more than 12.4 g/cm3.
(71) According to the embodiment, the Ar concentration in the Rh layer included in the p-side contact electrode **30** is smaller than the Ar concentration in the Rh layer **52** included in the p-side current diffusion layer **34** and is less than 1×10.sup.18/cm.sup.3 so that the film quality of the Rh layer included in the p-side contact electrode **30** can be improved. This allows the p-side contact electrode **30** to function as a highly efficient reflection electrode.
(72) According to the embodiment, inclusion of the Rh layer **52***b* in the p-side current diffusion layer **34** can increase the quality of sealing of the p-side contact electrode **30**. Similarly, inclusion of the Rh layer **70***b* in the n-side current diffusion layer **36** can increase the quality of sealing of the n-side contact electrode **32**. Further, the film density of the Rh layers **52***b*, **70***b* included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** is equal to or more than 12.0 g/cm.sup.3 so that the Rh layers **52***b*, **70***b* included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** can be configured to be dense to increase the quality of sealing even further.
(73) According to the embodiment, formation of the Rh layer included in the p-side contact electrode **30** by deposition can suppress the damage done to the upper surface **28***a* of the p-type semiconductor layer **28** when the Rh layer is formed and lower and the contact resistance of the p-side contact electrode **30** more successfully than sputtering. Meanwhile, formation of the Rh layer **52***b* included in the p-side current diffusion layer **34** by sputtering can increase adhesion to the p-side contact electrode **30** and suppress exfoliation of the p-side current diffusion layer **34** more successfully than deposition.
(74) According to the embodiment, formation of the Rh layer included in the p-side contact electrode **30** by deposition can prevent mixture of the Ar gas used in sputtering and improve the film quality of the Rh layer as compared with sputtering.
(75) According to the embodiment, annealing the p-side contact electrode **30** at a temperature equal to or more than 500° C. and equal to or less than 650° C. can lower the contact resistance of the p-side contact electrode **30** and improve the film density of the Rh layer included in the p-side contact electrode **30** as compared to the levels before the annealing.
(76) According to the embodiment, heating the p-side current diffusion layer **34** and the n-side current diffusion layer **36** at a temperature equal to or more than 200° C. and equal to or less than 400° C. can increase the film density of the Rh layers **52***b*, **70***b* included in the p-side current diffusion layer **34** and the n-side current diffusion layer **36** as compared to the level before the heating. This can increase the quality of sealing of the p-side current diffusion layer **34** and the n-side current diffusion layer **36**.
(77) Described above is an explanation based on an exemplary embodiment. The embodiment is intended to be illustrative only and it will be understood by those skilled in the art that various design changes are possible and various modifications are possible and that such modifications are also within the scope of the present invention.
(78) A description will be given below of some aspects of the present invention.
(79) A semiconductor light-emitting element according to the first aspect includes: an n-type semiconductor layer made of an n-type AlGaN-based semiconductor material; an active layer provided on the n-type semiconductor layer and made of an AlGaN-based semiconductor material; a p-type semiconductor layer provided on the active layer; a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively, wherein an Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer. According to the first aspect, the quality of sealing of the p-side contact electrode can be increased by covering the upper surface and the side surface of the p-side contact electrode, which includes the Rh layer, by

the p-side current diffusion layer, which includes the Rh layer. Further, the film quality of the p-side contact electrode can be improved, and the p-side contact electrode can be allowed to function as a highly efficient reflection electrode by configuring the Ar concentration to be small.
(80) The second aspect of the present invention relates to the semiconductor light-emitting element according to the first aspect, wherein the Ar concentration in the Rh layer included in the p-side contact electrode is less than 1×10.sup.18/cm.sup.3, and the Ar concentration in the Rh layer included in the p-side current diffusion layer is equal to or more than 1×10.sup.18/cm.sup.3. According to the second aspect, the film quality of the p-side contact electrode can be improved, and the p-side contact electrode can be allowed to function as a highly efficient reflection electrode.

Claims

1. A semiconductor light-emitting element comprising: an n-type semiconductor layer made of an n-type AlGaN-based semiconductor material; an active layer provided on the n-type semiconductor layer and made of an AlGaN-based semiconductor material; a p-type semiconductor layer provided on the active layer; a p-side contact electrode that includes a Rh layer in contact with an upper surface of the p-type semiconductor layer; and a p-side current diffusion layer that is in contact with an upper surface and a side surface of the p-side contact electrode and includes a TiN layer, a Ti layer, a Rh layer, and a TiN layer stacked successively, wherein an Ar concentration in the Rh layer included in the p-side contact electrode is smaller than an Ar concentration in the Rh layer included in the p-side current diffusion layer.
2. The semiconductor light-emitting element according to claim 1, wherein the Ar concentration in the Rh layer included in the p-side contact electrode is less than 1×10.sup.18/cm.sup.3, and the Ar concentration in the Rh layer included in the p-side current diffusion layer is equal to or more than 1×10.sup.18/cm.sup.3.